Exploratory Data Analysis (EDA)

Retail Store Sales Dataset

Prepared by: Tushar Sharma

Week 2 Internship Task | Python • pandas • Matplotlib • SciPy

Objective

Explore retail transaction data through descriptive statistics, visualizations, time-based summaries, and an initial hypothesis test. The report emphasizes reproducibility and transparent treatment of missing data.

Dataset and scope

The analysis uses the cleaned Week 1 dataset containing 12,575 rows and 11 columns. Transaction dates range from 2022-01-01 to 2025-01-18. Monetary values are described as recorded spend; missing Total Spent values are excluded from spend calculations.

1. Executive Summary

This EDA examines transaction completeness, spend distribution, category performance, quantity-spend relationships, payment methods, locations, and monthly/yearly recorded spend. A Welch two-sample t-test compares mean recorded transaction spend for rows marked as discounted versus not discounted.

Metric	Finding
Dataset size	12,575 rows × 11 columns
Recorded Total Spent	11,971 non-missing values
Mean / median spend	129.65 / 108.50
Missing Item	1,213
Missing Quantity / Total Spent	604 / 604
Missing Discount Applied	4,199
Exact duplicate rows	0
Hypothesis test p-value	0.799

2. Methodology

The cleaned dataset was loaded with pandas. Transaction Date was parsed as datetime, Discount Applied was represented as a nullable Boolean, and numeric fields were converted to numeric types. The analysis reviewed dimensions, data types, missingness, duplicates, descriptive statistics, category and location summaries, payment counts, distributions, correlations, and time aggregation. Missing values were not filled merely to make the dataset complete.

3. Data Quality Overview

Column	Missing	Approx. %
Discount Applied	4,199	33.4%
Item	1,213	9.6%
Total Spent	604	4.8%
Quantity	604	4.8%

There are no exact duplicate rows or duplicate Transaction IDs in this dataset. Item, Quantity, Total Spent, and Discount Applied contain missing values. Item names and discount status cannot be safely inferred from every row, so unknowns are retained. Analyses of spend exclude missing Total Spent values.

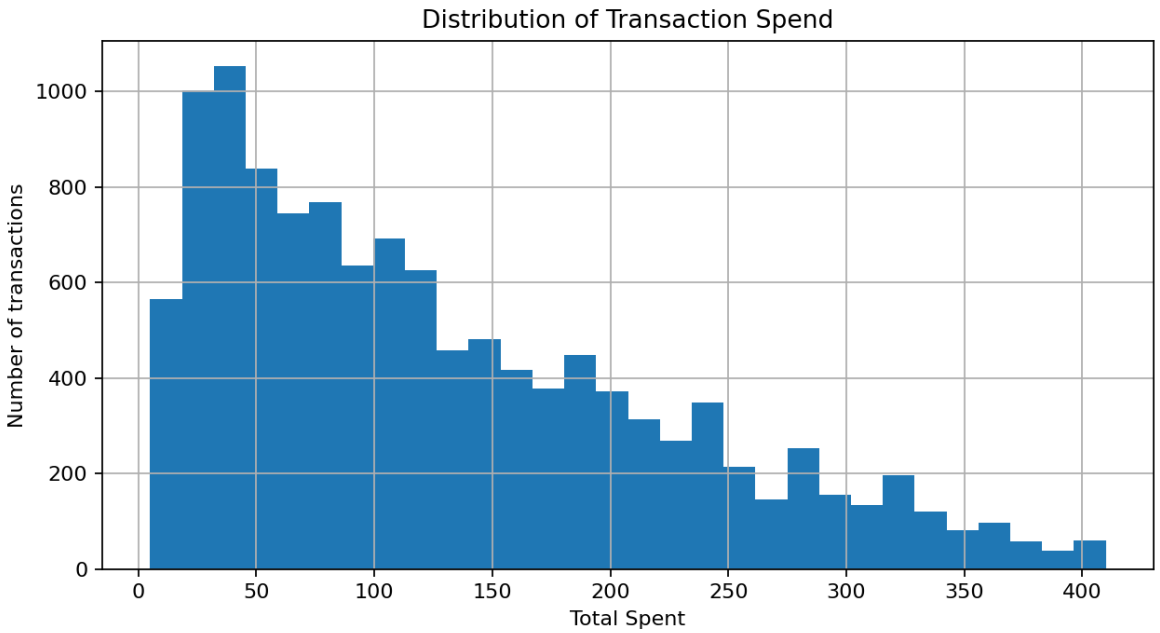
4. Descriptive Statistics

Variable	Count	Mean	Std. dev.	Min	Median	Max
Price Per Unit	12575	23.37	10.75	5.00	23.00	41.00
Quantity	11971	5.54	2.86	1.00	6.00	10.00
Total Spent	11971	129.65	94.75	5.00	108.50	410.00

Recorded Total Spent has mean 129.65 and median 108.50. The mean is higher than the median, consistent with a right-skewed distribution where some larger transactions pull the average upward. Spend varies considerably (standard deviation 94.75).

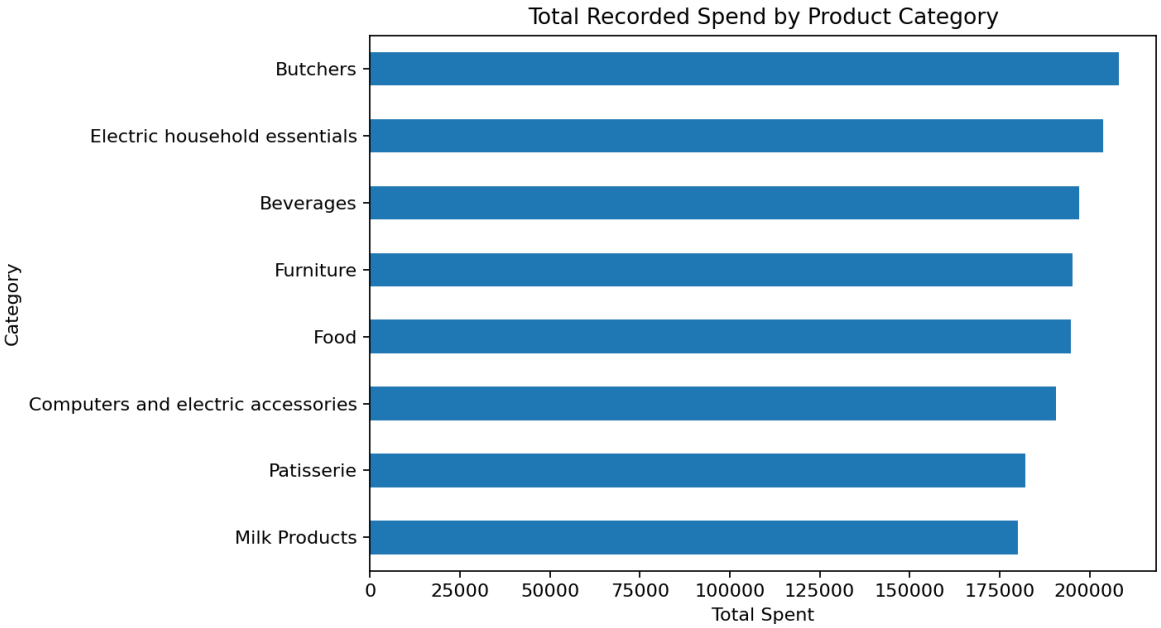
5. Visual Findings

5.1 Distribution of transaction spend



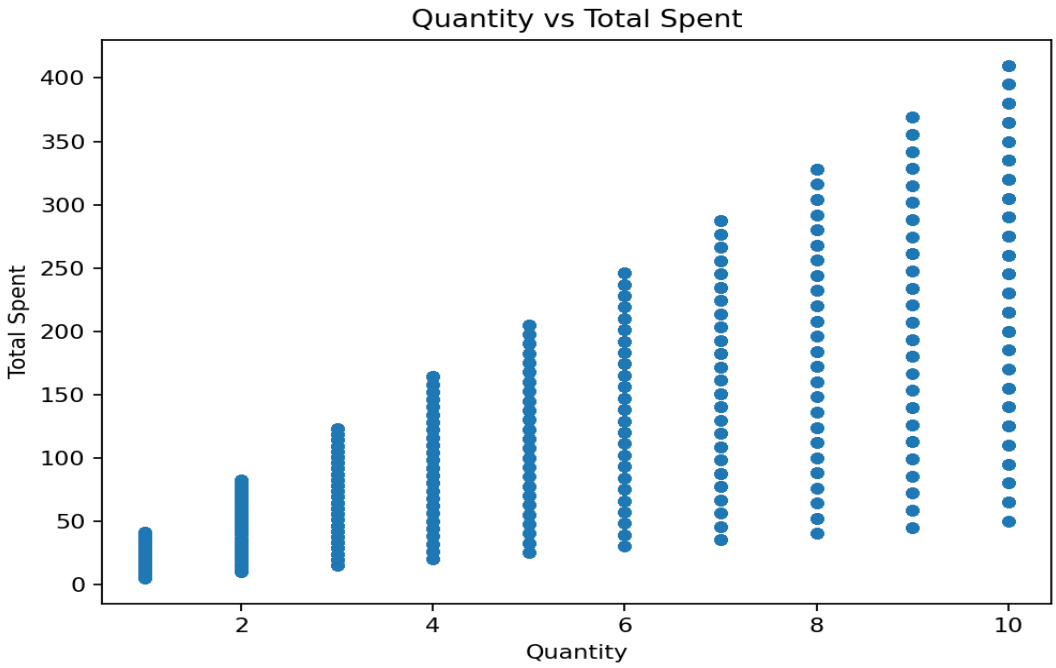
The histogram shows the spread of recorded transaction totals. The mean being above the median supports the interpretation that the distribution is not perfectly symmetric. Missing totals are excluded.

5.2 Recorded spend by category



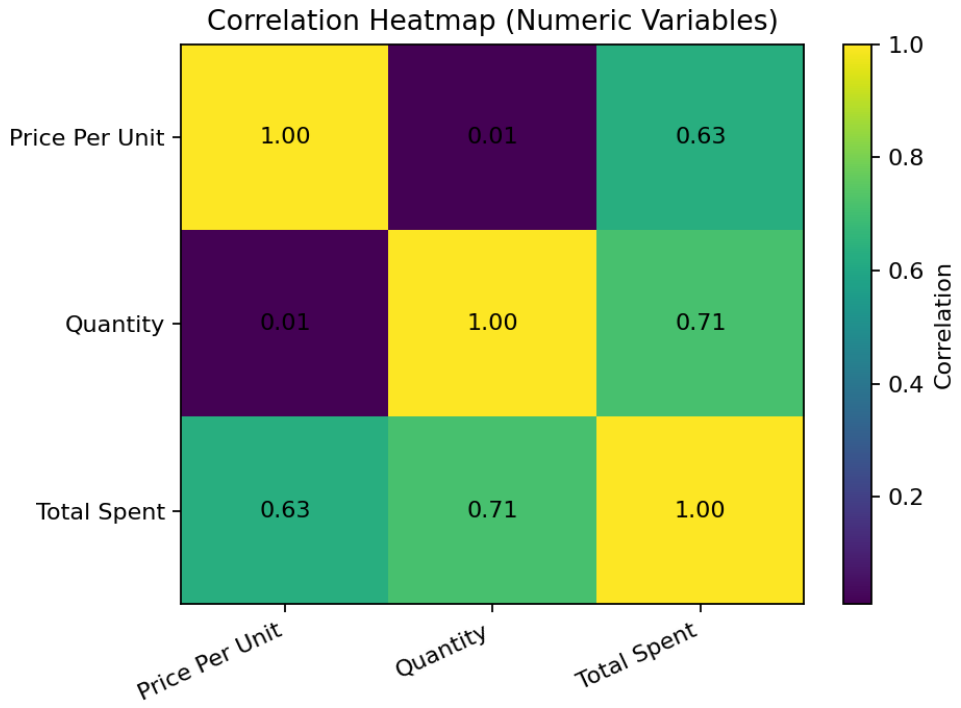
By recorded total spend, Butchers ranks highest (208,118.00) and Milk Products ranks lowest (180,112.00). This is total recorded spend, not profit; product costs and missing transactions are not accounted for.

5.3 Quantity versus spend



Quantity and Total Spent are expected to move together because transaction total is linked to unit price and quantity. The association is partly mathematical by construction and should not be interpreted as an independent causal discovery.

5.4 Correlation heat map



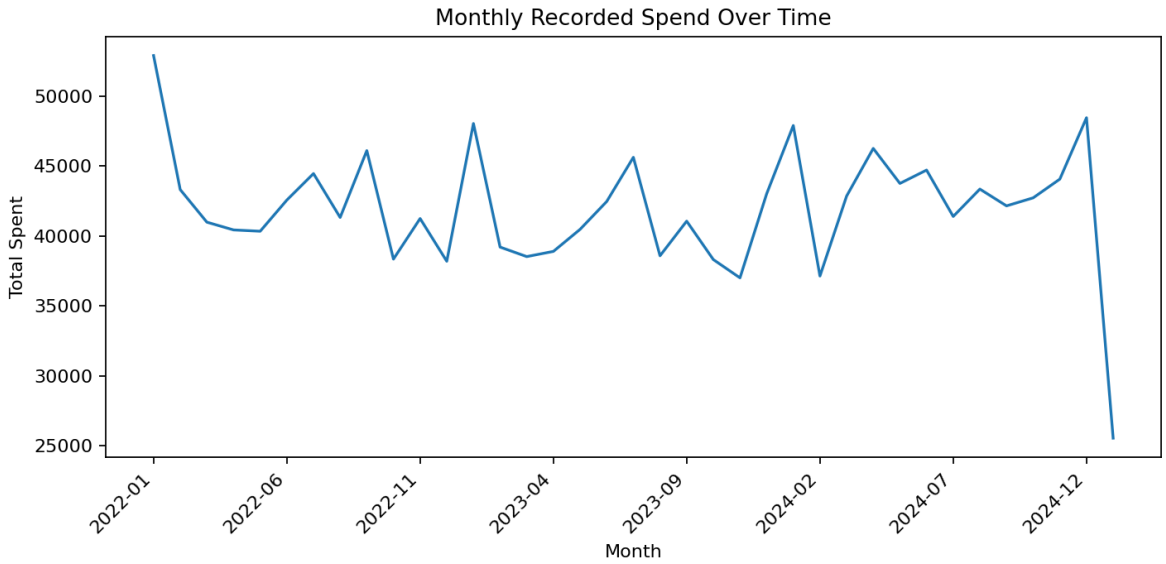
Correlation summarizes linear association between numeric variables. The relationship between Quantity and Total Spent is structurally expected. Correlation alone does not establish causation.

5.5 Discount status comparison



Recorded mean spend is 130.49 for transactions marked with a discount (n=4,019) and 129.95 for those marked without a discount (n=3,964). The medians are 109.50 and 109.50, respectively.

5.6 Monthly recorded spend



The timeline provides a view of monthly recorded spend. The dataset includes dates through 18 January 2025; therefore 2025 is only a partial year and should not be directly compared with full years without normalization.

6. Initial Hypothesis Test

Question: Is mean recorded transaction spend different between transactions marked as discounted and transactions marked as not discounted?

Null hypothesis (H■): The two population means are equal. **Alternative hypothesis (H■):** The two population means differ. Significance level $\alpha = 0.05$.

Welch's t-test result: t = 0.254, p = 0.799. Since p is greater than 0.05, we fail to reject the null hypothesis. This sample does not provide statistically significant evidence of a difference in mean recorded spend between the two recorded groups.

This result does not prove the means are identical and does not establish a causal effect of discounts. The discount field has substantial missingness (about one-third of rows), and unknown discount statuses were excluded. That missingness could bias the comparison.

7. Other Business Summaries

Location	Known spend rows	Recorded total	Average
Online	6,068	791,401.00	130.42
In-store	5,903	760,670.00	128.86

Online has a slightly higher recorded total spend than In-store in this dataset. Because this is a descriptive comparison and missing values remain, it should not be treated as proof of a meaningful business advantage.

Payment method counts

Payment Method	Transactions	Share of all rows
Cash	4,310	34.3%
Digital Wallet	4,144	33.0%
Credit Card	4,121	32.8%

Cash is the most frequently recorded payment method, followed by Digital Wallet and Credit Card. These counts describe transaction frequency rather than customer-level preference.

8. Limitations and Assumptions

• The dataset is a structured retail practice dataset; results should not automatically be generalized to a real retailer.
• Missing values are retained when they cannot be inferred safely. Most spend metrics use only rows with known Total Spent.

• Recorded total spend is not profit, and no cost data are available.
• The discount comparison excludes unknown discount statuses and may be affected by non-random missingness.
• The latest year is partial, so annual totals are not directly comparable.
• Statistical significance is not the same as business importance or causation.

9. Conclusion

The EDA summarizes the dataset with descriptive statistics, visualizations, category/location/payment comparisons, temporal aggregation, and an initial hypothesis test. Butchers has the highest recorded category spend; the recorded spend distribution has a mean above its median; and the discount-group t-test is not statistically significant at the 5% level. The most important caveat is data completeness: several fields contain unknown values. The notebook documents the analysis so it can be rerun and extended for later modeling tasks.

10. Reproducibility

Open `Week2_Retail_Sales_EDA.ipynb` in Jupyter Notebook or VS Code. Keep `retail_store_sales_cleaned.csv` beside the notebook, install pandas, NumPy, Matplotlib, and SciPy if needed, then run all cells from top to bottom. Figures are saved in the `figures` folder for the report.